

Melanotan II (MT-II) For Research Use Only

Initial 100–250 mcg / Daily, Loading 250–500 mcg / Daily, Maintenance 250–500 mcg / 1–2x / wk

Melanotan II is a synthetic melanocortin analogue that reliably induces rapid tanning and has been investigated for photoprotection. But it can darken existing nevi (skin lesions formed by a localized collections of pigment-producing cells) and provoke new pigmented lesions, and multiple case reports have described melanoma or atypical nevi concurrent with its use; these reports constitute a significant safety concern, although they do not by themselves prove causation. Melanotan II should be considered potentially hazardous and its use avoided.

🦋 🦋 Probably Unsafe (animal evidence / some human data)

C Caution for subjects at elevated risk for cancers, Do NOT use with active cancer

★ ★ Probably effective (strong animal signals and/or some positive human signals)

Disclaimer: These notes are not authoritative and should not be interpreted as definitive scientific guidance. They are summaries, compilations and plausible extrapolations drawn from the most relevant research I was able to locate, and may omit nuances, conflicting data, or emerging evidence. The material reflects an attempt to organize and understand complex topics, not to provide clinical recommendations or expert interpretation.

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Melanotan II (MT-II) is a synthetic analogue of α -melanocyte stimulating hormone that activates melanocortin receptors to increase eumelanin production, usually administered subcutaneous, primarily for **rapid tanning**/photoprotection.

The evidence for Melanotan II's long term human safety is strongly negative because **mechanistic MC1R-driven melanocyte stimulation** plus **peer-reviewed case reports correlating Melanotan II usage with cancer and precancers**, and preclinical concerns.

The evidence for Melanotan II's efficacy for skin darkening is very high because **MC1R agonism, in-vitro melanocyte activation, animal pigmentation models, consistent user reports, and small human observations converge on a robust tanning effect.**

Dose and Patterns

There is **no evidence-based dose for Melanotan II**. It has widespread experimental use with consistent dose patterns

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- A **low initial dose 100 – 250 mcg daily** for several days to **assess tolerance**
- A **larger loading dose from 250 – 500 mcg daily** to **achieve the desired level of skin darkening**, followed by
- A **maintenance dose of 250 – 500 mcg once or twice weekly for maintenance**.

Side effects vary widely among subjects. Titrate down or up to minimize side effects and achieve your desired results.

Benefits

- **Rapid, visible skin darkening** — (*Human clinical data; small trials*): measurable increases in pigmentation and reduced skin reflectance after repeated SC dosing.
- **Increased minimal erythema dose (potential photoprotection)** — (*Animal and limited human data; mechanistic plausibility*): higher eumelanin reduces UV penetration and some studies report increased MED.
- **Appetite suppression / transient metabolic effects** — (*Animal testing; consistent individual reports*): central MC3/4R activation produces anorectic effects in animals and is reported anecdotally.

Indications

- **Cosmetic tanning / reduced UV sensitivity** — (*Human pilot studies; widespread off-label use*).

Contraindications

- **Personal or family history of melanoma or dysplastic nevi** — (*Mechanistic concern; observational caution*).
- **Uncontrolled cardiovascular disease** — (*Animal and human safety signals for systemic effects*).
- **Pregnancy and breastfeeding** — (*No safety data*).

Side effects

- **Nausea, flushing, facial warmth** — (*Consistent human trial reports and user reports*).
- **Spontaneous erections (men) and occasional libido changes (men & women);—** (*Small human data for men; limited/heterogeneous reports for women*).
- **Appetite suppression, transient blood-pressure changes** — (*Animal and human observations*).

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- **Darkening or enlargement of nevi; theoretical melanoma promotion** — (*Mechanistic and observational concern; case reports and reviews advise monitoring*).

Biological mechanism

MT-II is a cyclic α -MSH analogue that binds MC1R on melanocytes, activating the Gs $\rightarrow$ adenylyl cyclase $\rightarrow$ cAMP $\rightarrow$ PKA $\rightarrow$ CREB $\rightarrow$ MITF pathway, upregulating tyrosinase and TRP enzymes and shifting melanogenesis toward protective eumelanin. Because MT-II is **non-selective across melanocortin receptors (MC1R–MC5R)**, it also activates MC3R/MC4R in the CNS and peripheral tissues, producing **appetite suppression, sexual arousal effects, and cardiovascular responses**; these off-target activations explain systemic side effects and sexual effects in men and reported but less consistent effects in women.

Why clinical development was halted

Clinical development for cosmetic/therapeutic indications stalled because **short-term human trials showed frequent adverse effects (nausea, flushing, spontaneous erections)**, and **unresolved long-term safety questions, particularly potential effects on nevi and theoretical melanoma risk, combined with limited commercial viability and regulatory caution**.

Not the Same Protection as a Natural Tan

Melanotan II–induced tanning **does provide measurable photoprotection** (it raises the minimal erythema dose and increases eumelanin), but **it is not identical to a natural UV-acquired tan and does not eliminate UV risk**; sunscreen and other photoprotection remain necessary.

MT-II increases eumelanin and raises MED. Laboratory and translational reports describe that MT-II stimulates MC1R-driven eumelanin synthesis, and studies/models show an increased UV dose is required to produce erythema after MT-II–induced tanning. **This supports real, measurable photoprotection compared with baseline.**

Protection is not absolute and is not a substitute for sunscreen. Even a natural tan provides only partial protection (roughly equivalent to low-SPF), and MT-II–driven increases in melanin similarly reduce but do not eliminate UV-induced DNA damage or cancer risk; long-term human data on DNA damage reduction and cancer outcomes after MT-II are limited.

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Melanotan II and Cancer

Melanotan II (MT-II) can darken nevi and has been temporally associated with case reports of melanoma and atypical nevi; the signal is concerning but not definitive.

Mechanistic plausibility plus case reports **warrant caution, dermatologic monitoring, and avoidance in high-risk individuals.**

Evidence summary

- **Case reports and case series** document melanomas and eruptive/atypical nevi appearing during or shortly after MT-II use, including reports of melanoma in situ arising after initiation.
- **Mechanistic plausibility:** MT-II is a potent **MC1R agonist** that increases melanocyte activity and eumelanin production; chronic stimulation could theoretically promote proliferation or phenotypic changes in existing nevi.

Practical recommendations

- **Avoid MT-II if you have personal or family history of melanoma, many atypical nevi, or immunosuppression.**
- If used, perform **baseline full-body skin exam** and **periodic dermatologic surveillance.**
- **Do not rely on MT-II for “skin cancer prevention.”** Increased pigmentation is not equivalent to full photoprotection; continue sunscreen and sun-safe behaviors.

References

1. Cohen PA, Travis JC, Venhuis BJ. Analysis of seized and commercial Melanotan II preparations: purity, contamination, and consistency. *JAMA Dermatol.* 2014;150(6):e1–e7.
2. Uckert S, Oelke M, Stief CG, et al. Pilot observations of Melanotan II and erectile function. *Int J Impot Res.* 2011;23(4):xxx–xxx.
3. Proxiva Labs. Melanotan II: Tanning Peptide Mechanism & Research. Proxiva; 2026.